$$P_c = K\rho^{4/3} \qquad K = \frac{3}{10}\left(\frac{4\pi}{3}\right)^{1/3}\frac{Z^{4/3}e^2}{(m_p M_e)^{4/3}} \qquad \frac{\rho}{\mu_e} = Bx^3$$

$$z^2 = x^2 + 1$$

$$P_c(x) = K(B\mu_e x^3)^{4/3} = K(B\mu_e)^{4/3}x^4$$

$$\frac{1}{\rho}\frac{dP_c}{dr} = \frac{1}{\rho}\frac{dP_c}{dx}\frac{dx}{dz}\frac{dz}{dr} \qquad \frac{dx}{dz} = \frac{d\sqrt{z^2-1}}{dz} = \frac{z}{x}$$

$$= \frac{1}{B\mu_e x^3}\left(4K(B\mu_e)^{4/3}x^3\right)\left(\frac{z}{x}\right)\frac{dz}{dr}$$

$$= \frac{4K(B\mu_e)^{1/3}}{x}z\frac{dz}{dr} \qquad \frac{1}{\rho}\frac{dP_e}{dr} = \frac{8A}{B\mu_e}\frac{dz}{dr}$$

$$\frac{1}{r^2}\frac{d}{dr}\left(\frac{r^2}{\rho}\frac{d\rho}{dr}\right) = \frac{1}{r^2}\frac{d}{dr}\left(\frac{8A}{B\mu_e}r^2\frac{dz}{dr} + 4K(B\mu_e)^{1/3}r^2\frac{z}{x}\frac{dz}{dr}\right)$$

$$= -4\pi G B\mu_e x^3$$

$$\frac{d}{dr}\left(\left(\frac{8A}{B\mu_e} + 4K(B\mu_e)^{1/3}\frac{z}{x}\right)r^2\frac{dz}{dr}\right)$$

$$= \left(\frac{8A}{B\mu_e} + 4K(B\mu_e)^{1/3}\frac{z}{x}\right)\frac{d}{dr}\left(r^2\frac{dz}{dr}\right) + 4K(B\mu_e)^{1/3}\frac{d}{dr}\left(\frac{z}{x}\right)r^2\frac{dz}{dr}$$

$$\frac{d}{dr}\left(r^2\frac{dz}{dr}\right) = 2r\frac{dz}{dr} + r^2\frac{d^2z}{dr^2}$$

$$\frac{dx}{dr} = \frac{d\sqrt{z^2-1}}{dr}$$

$$= \frac{d\sqrt{z^2-1}}{d(z^2-1)}\frac{d(z^2-1)}{dr}$$

$$= \frac{1}{2\sqrt{z^2-1}}(2z)\frac{dz}{dr}$$

$$\frac{d}{dr}\left(\frac{z}{x}\right) = \frac{1}{x^2}\left(x\frac{dz}{dr} - z\frac{dx}{dr}\right)$$

$$= \frac{1}{x^2}\left(x\frac{dz}{dr} - \frac{z^2}{x}\frac{dz}{dr}\right)$$

$$= \frac{1}{x^2}\left(\frac{x^2-(x^2+1)}{x}\right)\frac{dz}{dr}$$

$$= -\frac{1}{x^3}\frac{dz}{dr}$$

$$\frac{1}{r^2}\left(\left(\frac{8A}{B\mu_e}+4\pi K(B\mu_e)^{1/3}\frac{z}{x}\right)\left(2r\frac{dz}{dr}+r^2\frac{d^2z}{dr^2}\right)-\frac{4\pi K(B\mu_e)^{1/3}r^2}{x^3}\left(\frac{dz}{dr}\right)^2\right)$$

$$=\left(\frac{8A}{B\mu_e}+4\pi K(B\mu_e)^{1/3}\frac{z}{x}\right)\left(\frac{2}{r}\frac{dz}{dr}+\frac{d^2z}{dr^2}\right)-\frac{4\pi K(B\mu_e)^{1/3}}{x^3}\left(\frac{dz}{dr}\right)^2\Bigg\}$$

$$=-4\pi G B\mu_e x^3$$

$$x=\sqrt{z^2-1}$$

$$Q\equiv\frac{z}{z_c}\qquad \xi\equiv\frac{r}{\alpha}\qquad \alpha\equiv\sqrt{\frac{2A}{\pi G}}\left(\frac{1}{B\mu_e z_c}\right)$$

$$\frac{d}{d(\alpha\xi)}\frac{d}{d(\alpha\xi)}(z_cQ)=\frac{z_c}{\alpha^2}\frac{d^2Q}{d\xi^2}$$

$$\left(\frac{8A}{B\mu_e}+\frac{4\pi K(B\mu_e)^{1/3}z_cQ}{\sqrt{z_c^2Q^2-1}}\right)\left(\frac{2}{\alpha\xi}\frac{d(z_cQ)}{d(\alpha\xi)}+\frac{z_c}{\alpha^2}\frac{d^2Q}{d\xi^2}\right)$$

$$-\frac{4\pi K(B\mu_e)^{1/3}z_c^2}{x^3}\frac{z_c}{\alpha^2}\left(\frac{dQ}{d\xi}\right)^2=-4\pi G B\mu_e(z^2-1)^{3/2}$$

$$\left(\frac{8A}{B\mu_e}+\frac{4\pi K(B\mu_e)^{1/3}z_cQ}{\sqrt{z_c^2Q^2-1}}\right)\left(\frac{2}{\xi}\frac{dQ}{d\xi}+\frac{d^2Q}{d\xi}\right)z_c-\frac{4\pi K(B\mu_e)^{1/3}z_c^2}{x^3}\left(\frac{dQ}{d\xi}\right)^2$$

$$=-4\pi G B\alpha^2\mu_e(z^2-1)^{3/2}=-4\pi G B\alpha^2\mu_e(z_c^2Q^2-1)^{3/2}$$

$$=-4\pi G B\alpha^2\mu_e z_c^3\left(Q^2-\frac{1}{z_c^2}\right)^{3/2}$$

$$\left(\frac{8A}{B\mu_e}+\frac{4\pi K(B\mu_e)^{1/3}z_cQ}{\sqrt{z_c^2Q^2-1}}\right)\left(\frac{2}{\xi}\frac{dQ}{d\xi}+\frac{d^2Q}{d\xi}\right)z_c-\frac{4\pi K(B\mu_e)^{1/3}z_c^2}{(z_c^2-1)^{3/2}}\left(\frac{dQ}{d\xi}\right)^2$$

$$=-4\pi G B\alpha^2\mu_e z_c^3\left(Q^2-\frac{1}{z_c^2}\right)^{3/2}$$

$$-4\ln G_B\left(\frac{2A}{\pi G}\right)\frac{1}{B^2\mu_e^2 z_c^2}\mu_e z_c^3\left(Q^2-\frac{1}{z_c^2}\right)^{3/2} = -\frac{4\pi(2A)}{B\mu_e} z_c\left(Q^2-\frac{1}{z_c^2}\right)^{3/2}$$

$$= -\frac{8A}{B\mu_e} z_c\left(Q^2-\frac{1}{z_c^2}\right)^{3/2}$$

$$\left(\frac{8A}{B\mu_e}+\frac{4K(B\mu_e)^{1/3}z_c Q}{\sqrt{z_c^2Q^2-1}}\right)\left(2\frac{dQ}{d\xi}+\xi\frac{d^2Q}{d\xi^2}\right)-\frac{4K(B\mu_e)^{1/3}}{(z_c^2Q^2-1)^{3/2}} z_c\left(\frac{dQ}{d\xi}\right)^2$$

$$= -\frac{8A}{B\mu_e}\left(Q^2-\frac{1}{z_c^2}\right)^{3/2} \qquad Q(0)=1 \quad Q'(0)=0$$

$$\left(\frac{8A}{B\mu_e}+\frac{4K(B\mu_e)^{1/3}z_c}{\sqrt{z_c^2-1}}\right)3Q'(0) = -\frac{8A}{B\mu_e}\left(1-\frac{1}{z_c^2}\right)^{3/2}$$

$$Q'(0) = -\frac{8A}{3B\mu_e}\left(1-\frac{1}{z_c^2}\right)^{3/2}\left(\frac{8A}{B\mu_e}+\frac{4K(B\mu_e)^{1/3}z_c}{\sqrt{z_c^2-1}}\right)^{-1}$$

$$Q(\xi) = Q(0)+Q'(0)\xi+\frac{Q''(0)}{2}\xi^2+\dots$$

$$= 1-\frac{4A}{3B\mu_e}\left(1-\frac{1}{z_c^2}\right)^{3/2}\left(\frac{8A}{B\mu_e}+\frac{4K(B\mu_e)^{1/3}z_c}{\sqrt{z_c^2-1}}\right)^{-1}\xi^2+\dots$$

$$Q(\xi) = Q(0)+Q'(0)\xi+\dots$$

$$= -\frac{8A}{3B\mu_e}\left(1-\frac{1}{z_c^2}\right)^{3/2}\left(\frac{8A}{B\mu_e}+\frac{4K(B\mu_e)^{1/3}z_c}{\sqrt{z_c^2-1}}\right)^{-1}\xi$$

$$m = 4\pi\alpha^3 B\mu_e z_c^3\int_0^{\xi_1}\xi^2\left(Q^2-\frac{1}{z_c^2}\right)^{3/2}d\xi$$

$$\rho = B\mu_e z_c^3\left(Q^2-\frac{1}{z_c^2}\right)^{3/2} \qquad m = 4\pi\int_0^r \rho r^2\,dr$$

$$r = \alpha\xi$$